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Based on the two-dimensional air resistance bridge crane anti-swing control research

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Abstract

Aiming at the anti-pendulum problem of bridge crane system, a two-dimensional model of bridge crane system under the action of damping force is established in this paper. Considering the influence of air resistance and friction between trolly and track in complex working environment, a novel air resistance model and friction model are introduced to establish a more accurate two-dimensional bridge crane model with variable rope length under the action of damping force.Based on the fuzzy adaptive PID controller, appropriate fuzzy rules are designed to realize the anti-pendulum positioning control of bridge crane.The simulation results show that compared with the multi-sliding mode controller, the controller designed in this paper has better control effect and strong robustness.

Keywords: bridge crane, fuzzy adaptive PID control, air resistance model, anti-swing positioning control

1. Introduction

As an underactuated system, external damping forces such as air resistance and friction force are often ignored in the process of building the model of bridge crane system, which makes the system modeling inaccurate and results in great difference between the simulation and the actual performance. Literature [1] proposed a linear wind model, and the size of wind parameters is related to the windward area. Literature [2] designed a simulator to predict and simulate wind power compensation for wind power interference. Literature [3] considers the two-dimensional wind disturbance of load in the two-dimensional bridge crane system, and adopts cosine and sine wind models in two

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directions respectively. However, the model constructed by the above method is difficult to be used in the actual controller design due to its complex structure and high coupling.

In terms of controller design, sliding mode variable structure control [4] and PID [5] control technology are widely applied in bridge crane system. However, sliding mode controller requires accurate system model and has high energy consumption and strong buffeting problems. In contrast, PID performance is more stable, the structure is simple and easy to achieve. However, the performance of traditional PID control is excessively dependent on the selection of parameters, while the optimization methods represented by particle swarm optimization [6] and neural network algorithm [7] are difficult to meet the requirements of engineering practice.

In order to solve the above problems, the air resistance and friction under complex environment are considered in the dynamic modeling process, and a new two-dimensional bridge crane system model is established. Based on this, the fuzzy adaptive PID controller is designed and applied to the anti-pendulum positioning control of the proposed 2-D bridge crane system. The validity and robustness of the proposed method are verified by comparing with the simulation curves of PID controller and multi-sliding mode controller.

2. Dynamic Model

2.1. System Introduction

As an underactuated nonlinear system dynamic model, the two-dimensional bridge crane is shown in FIG. 1.

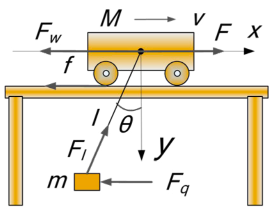

Fig. 1. Model of bridge trolley

In this paper, the dynamic equation of bridge crane system is obtained by using Lagrange equation:.

$$\begin{cases} (m+M)\ddot{x} - m\ddot{l}\sin\theta - ml\dot{\theta}^2\sin\theta - 2m\dot{l}\dot{\theta}\cos\theta - ml\ddot{\theta}\cos\theta = F - (F_w + f) \\ 2m\dot{l}\dot{\theta} + ml\ddot{\theta} - m\ddot{x}\cos\theta + mg\sin\theta = F_\theta \\ m\ddot{l} - m\ddot{x}\sin\theta - ml\dot{\theta}^2 - mg\cos\theta = F_l \end{cases} \tag{1}$$

where M, m and l respectively represent the mass of the trolley, the payload mass and the rope length; x and θ represents the horizontal displacement of the trolley and the swing angle of the load respectively; F_w, f and F_θ are the air quality of the trolley, the rail friction and the air quality of the payload. F and F_l are the driving force of the trolley and the lifting force of the rope, respectively.

In order to make the dynamic equation of the bridge crane system more in line with the actual engineering situation, the friction resistance and air resistance of the system are fully considered in the process of the system model establishment, and the model is improved. In the process of working, the friction between the bridge crane and the track objectively exists, and its magnitude is related to the trolley's moving speed and other factors. In addition, the direction of air resistance of a moving object is opposite to its motion direction, and its magnitude is related to its velocity, air density, object geometry, volume and other factors [9]. In practice, bridge cranes usually operate at a low speed in order to avoid excessive payload swing angle or accidents. Therefore, the low-speed motion air resistance and friction model are adopted in this paper, as shown in (4)

$$\begin{cases} f = f_{r0}\tanh(\dfrac{\dot{x}}{\sigma}) - k_r\left|\dot{x}\right|\dot{x} \\ F_w = ks_1\dot{x}^2 + ks_2(\dot{x} - \dot{l}\sin\theta - l\dot{\theta}\cos\theta)^2 \\ F_\theta = ks_2 l\left[(\dot{x} - \dot{l}\sin\theta - l\dot{\theta}\cos\theta)\cos\theta\right]^2 \end{cases} \tag{2}$$

where f_{r0}, σ and k_r are the coefficient of friction. k is the coefficient of air resistance consider of air density. s_1 and s_2 represent the windward area of the trolley moving direction and the windward area of the load moving direction respectively.

3. Control System Design

3.1. Principle of Fuzzy Adaptive PID Controller

PID controller is a common controller, which is often used for anti-swing control of trolley. Its basic structure is as follows:

$$F = K_p e + K_{ix}\int_0^t e d\tau + K_{dx} ec \tag{3}$$

where e and ec respectively represent the system deviation t and deviation rate, and K_{pq}, K_{iq} and K_{dq} are all normal control gains. Combined with their characteristics, the fuzzy adaptive PID control algorithm is derived. The control principle diagram of the fuzzy adaptive PID controller is as follows, as shown in Figure 2:

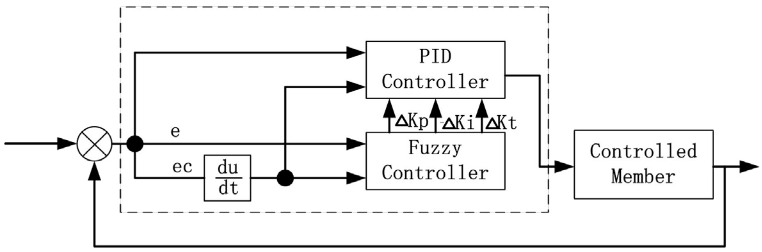

Fig. 2. Schematic diagram of Fuzzy Adaptive PID Controller

According to Figure 2, the specific working principle of the fuzzy adaptive PID controller is as follows: taking $e(t)$ and $ec(t)$ as input, the fuzzy reasoning is carried out by using the fuzzy control rules, and the PID parameters K_p, K_i and K_d are adjusted online to meet the self-tuning requirements of $e(t)$ and $ec(t)$ for PID parameters at different times.

3.2. Fuzzy Adaptive PID Controller Design

According to the model analysis of the anti-pendulum system of the bridge trolley, it can be known that it should contain two PID controllers, which are used to control the position of the trolley and the swing angle of the payload respectively. Combined with the controller principle mentioned above, the values of each variable of the two controllers are given as shown in Table 1.

Table 1. Design Table of Fuzzy Adaptive PID Controller

Variable		e	ec	$\Delta\boldsymbol{K}_p$	$\Delta\boldsymbol{K}_i$	$\Delta\boldsymbol{K}_d$
Language Variable		E	EC	$\Delta\boldsymbol{K}_p$	$\Delta\boldsymbol{K}_i$	$\Delta\boldsymbol{K}_d$
Base Field	$\boldsymbol{x}$	[-3 3]	[-0.5 0.5]	[-3 3]	[-0.1 0.1]	[-3 3]
	$\boldsymbol{\theta}$	[-0.03 0.03]	[-0.05 0.05]	[-0.6 0.6]	[-0.1 0.1]	[-0.6 0.6]
	$\boldsymbol{l}$	[-1 1]	[-1 1]	[-3 3]	[-0.1 0.1]	[-3 3]
Fuzzy Field	$\boldsymbol{x}$	[-3 3]	[-3 3]	[-3 3]	[-0.6 0.6]	[-3 3]
	$\boldsymbol{\theta}$	[-1 1]	[-1 1]	[-0.1 0.1]	[-0.1 0.1]	[-0.6 0.6]
	$\boldsymbol{l}$	[-3 3]	[-3 3]	[-3 3]	[-3 3]	[-3 3]
Quantifying Factors	$\boldsymbol{x}$	1	6	1	1/6	1
	$\boldsymbol{\theta}$	33	20	6	1	1
	$\boldsymbol{l}$	1/3	1/3	1	1/30	1
Fuzzy Subset		{*NB,NM,NS,Z0,PS,PM,PB*}				

Table 2. Table of Fuzzy Control Rules of ΔK_p, ΔK_i and ΔK_d

$\Delta\boldsymbol{K}_p/\Delta\boldsymbol{K}\Delta/\boldsymbol{K}_d$		EC						
		NB	NM	NS	Z0	PS	PM	PB
E	NB	PB/NB/PS	PB/NB/NS	PB/NM/NB	PM/NM/NB	PS/NS/NB	Z0/Z0/NM	Z0/Z0/PS
	NM	PB/NB/PS	PB/NB/NS	PB/NM/NB	PS/NS/NM	PS/NS/NM	Z0/Z0/NS	NS/Z0/Z0
	NS	PM/NM/Z0	PM/NM/NS	PS/NS/NM	PS/NS/NM	Z0/Z0/NM	NS/PS/NS	NS/PS/Z0
	Z0	PM/NM/Z0	PM/NM/NS	PS/NS/NS	Z0/Z0/NS	NS/PS/NS	NM/PM/Z0	NM/PM/Z0
	PS	PS/NM/Z0	PS/NS/Z0	Z0/Z0/Z0	NS/PS/Z0	NS/PM/Z0	NM/PM/Z0	NM/PB/Z0
	PM	PS/Z0/PB	Z0/Z0/NS	NS/PS/PS	NM/PS/PS	NM/PM/PS	NB/PB/PS	NB/PB/PB
	PB	Z0/Z0/PB	Z0/Z0/PM	NM/PS/PM	NM/PM/PM	NM/PM/PS	NB/PB/PS	NB/PB/PB

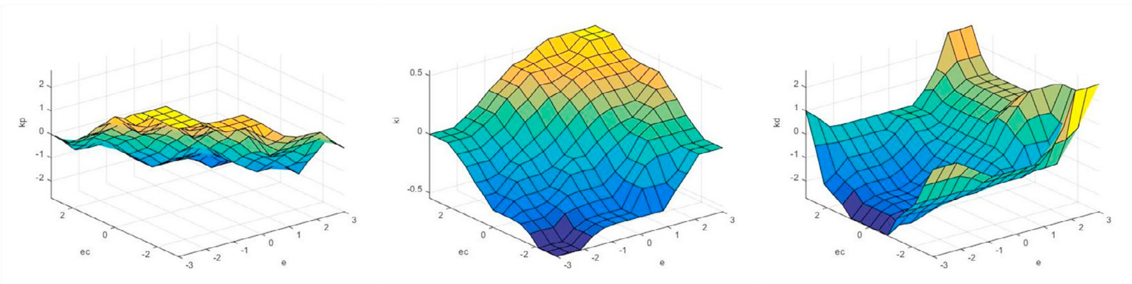

Fig. 3. Output Characteristic Surface of Position Fuzzy Reasoning. (a) K_p; (b) K_i; (c) K_d

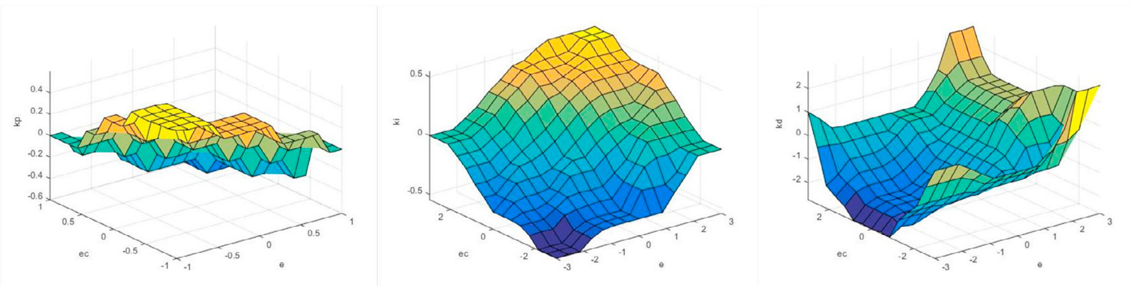

Fig. 4. Output Characteristic Surface of swing angle Fuzzy Reasoning. (a) K_p; (b) K_i; (c) K_d

The correction of the three parameters of the PID controller obtained by fuzzy control can obtain the real-time control parameters through Formula (9), and the obtained parameters can be used in the controller formula (7) and (8) to achieve better control results.

$$\begin{cases} K_p = K_{p0} + \Delta K_p \\ K_i = K_{i0} + \Delta K_i \\ K_\mathrm{d} = K_{d0} + \Delta K_d \end{cases} \tag{4}$$

4. Simulation And Experiment

To verify the effectiveness of the proposed model and controller, MATLAB/Simulink is used to simulate the bridge trolley model and its control algorithm. In order to make the simulation results closer to the actual working conditions, according to the actual bridge crane parameters combined with the similarity reasoning method, the model parameters are setting as follow: M=10kg, m=5kg, l_0=3m, g=9.78m/s^{2,} s_1=0.002m^2, s_2=0.004m^2, k=1kg/m^{3,} f_{r0}=0.2, σ=0.01, k_r=-0.05

The controller designed in this paper is compared with PID control and multi-sliding mode control in literature [4] for simulation experiment and comparative analysis. Due to space limitations, the expression of the contrast controller is not given here. After full debugging, the control gain of the method in this paper is selected as K_{px0}=2.4, K_{ix0}=0.001, K_{dx0}=16.3, $K_{p\theta0}$=22, $K_{i\theta0}$=0, $K_{d\theta0}$=6, K_{pl0}=30, K_{il0}=0 and K_{dl0}=20。

Experiment 1: According to the above parameters, we carried out simulation experiments without interference, and the simulation result as shown in Figure 5.

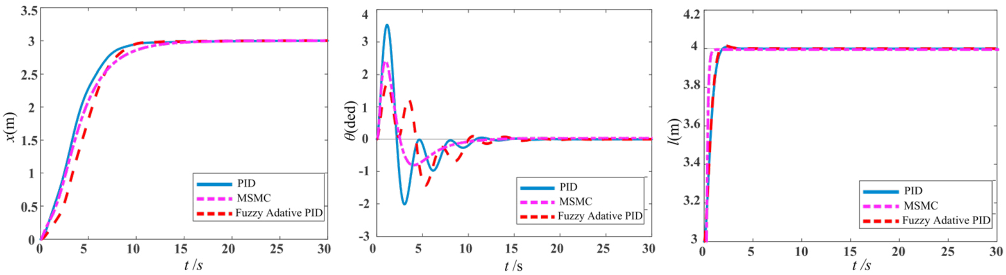

Fig. 5. Simulation Curve of experiment 1. (a) Displacement of trolley; (b) Swing angle; (c) Rope length

Analysis of data from FIG. 5 , the conclusion as follow: 1) Shown figure 5-a, the effect of the three controllers on the control displacement curve is similar, among which the convergence time of the fuzzy PID controller and common PID controller is similar (around 10s), but the displacement arrival time of the sliding mode controller is 14 seconds. 2) In figure 5-b, 3 kinds of controller all can eliminate residual swing angle in about 15 seconds, but the maximum swing angle of the payload using the fuzzy PID controller is 1.6 °, compared with the PID controller(3.5 °) and sliding mode controller(2.3 °) has great advantage; 3) As given in Figure 5-c. As the simulation is a process from 3m to 4m, the control time is relatively short, and the convergence time of PID and fuzzy PID controller is 3 s. The convergence time of sliding mode control is only 1s, but this may lead to higher instantaneous power.

Experiment 2: The working environment of bridge trolley is complex, so the controller is required to have strong robustness to external disturbance. In order to test the anti-interference capability of the three controllers mentioned above, we imposes a transient disturbance on the load to increase the payload swing angle by 0.5° at 20s which time the system is in a stable convergence state. The control results of each controller are shown in FIG 7.

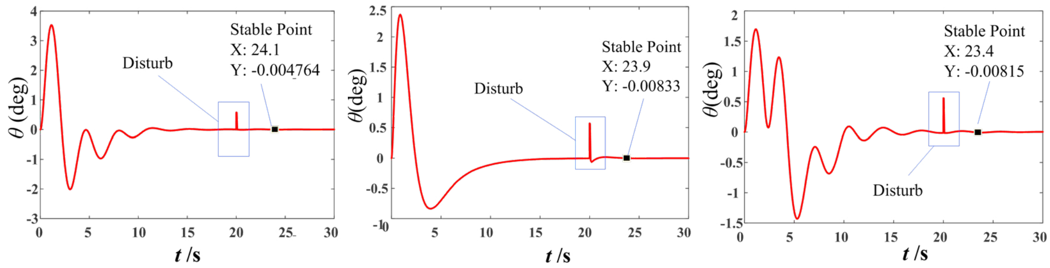

Fig. 6. Simulation curve of experiment 2. (a) PID controller; (b) Silding mode controller; (c) Fuzzy adaptive PID controller

By analyzing Figure 6, we can come to the conclusion that, after the disturbance of about 0.5° is increased in the 20s, the load swing Angle of the trolley produces jitter to different degrees under the control of three controllers, and successively converges back to 0 in a limited time. Among them, the fuzzy PID controller can be restored to the equilibrium state within 3.4s, while the time spent by the PID controller and the sliding mode variable structure controller is 4.1s and 3.9s respectively. The results show that the proposed fuzzy PID controller has better robustness than the other two controllers.

5. Conclusion

In order to solve the problem of anti-pendulum control of bridge trolley system in complex working environment, a two-dimensional model of bridge trolley system based on damping force is proposed in this paper. At the same time, a fuzzy adaptive PID controller is proposed for the anti-pendulum positioning control of this model.By comparison with the simulation results of PID and a sliding mode controller, a fuzzy adaptive PID controller has better positioning anti-swing ability and has strong robustness to external disturbances, is expected to be applied to improve the production efficiency of trolley actually, at the same time, under the complex environment modeling method can expand the application of the underactuated system to other complex environment.At present, the controller has only done simulation experiments, and the next step will be to carry out further research based on hardware experiment.

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